

PRST 8/09/2020

Data set 1
(10, 11, 12, 13)

Var 1.25

Data set 2
(1001, 1002, 1003, 1004)

Var = 1.25

~~10001, 10002, 10003, 10004~~

Coefficient of variation

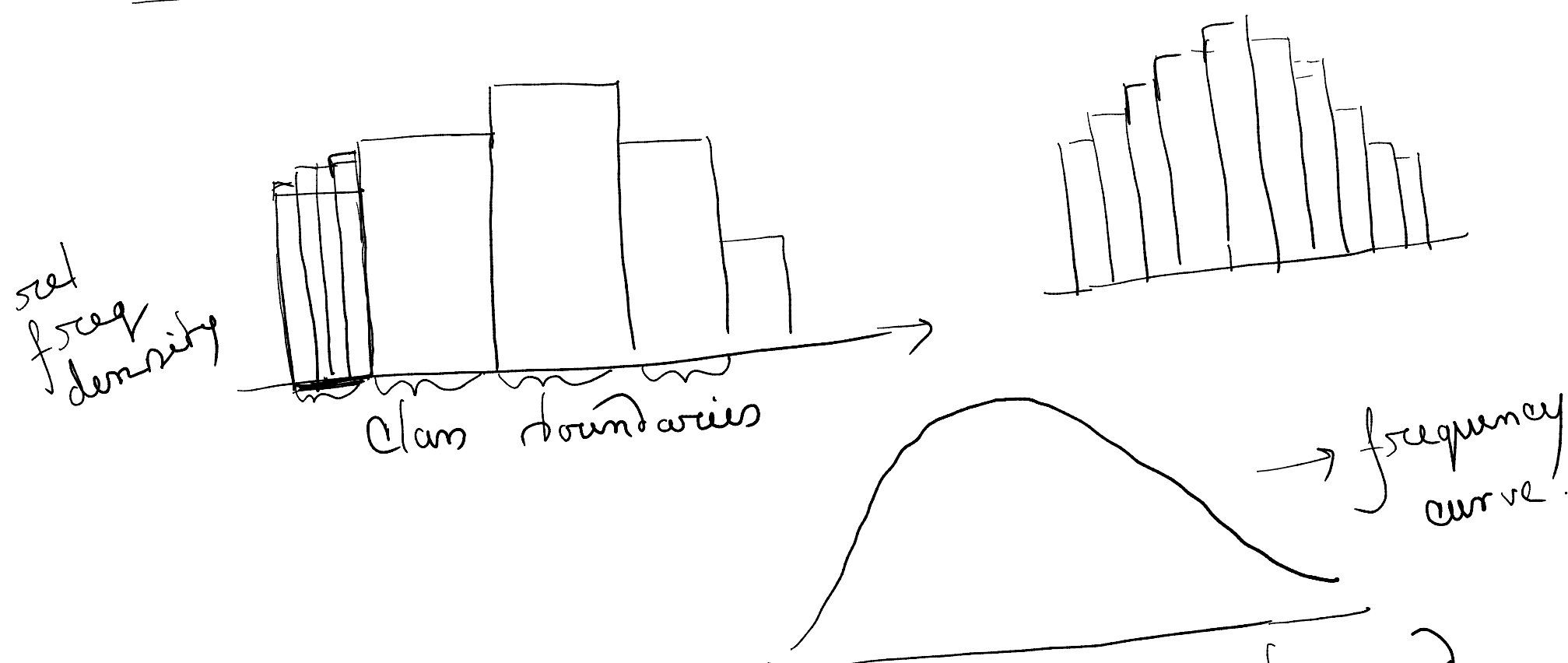
$$CV = 100 \times \frac{SD}{\bar{x}}$$

$CV_1 = 9.7\%$
Data set 1

$$SD = \sqrt{Var}$$

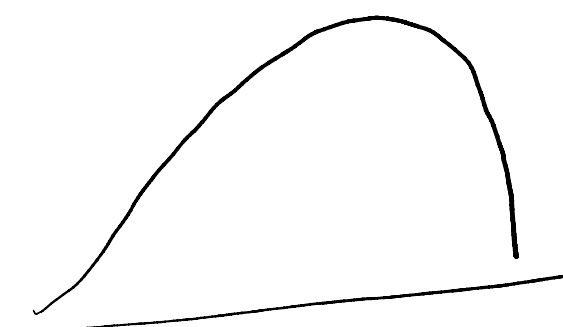
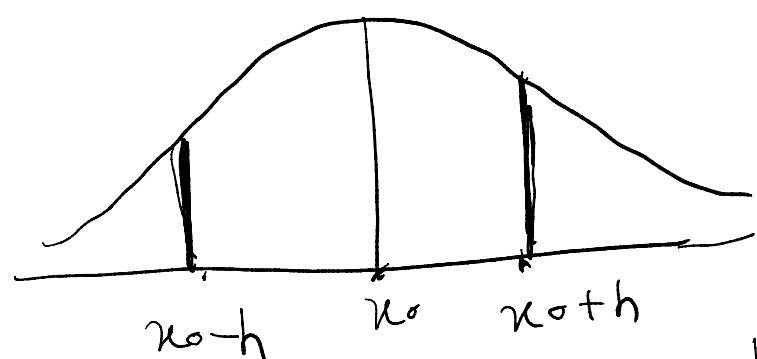
$CV_2 = 1.1\%$
Data set 2

- ① Data sets have different measurement units, and/or
 - ② Data sets have different means.
- Use CV to compare dispersions among these data sets.

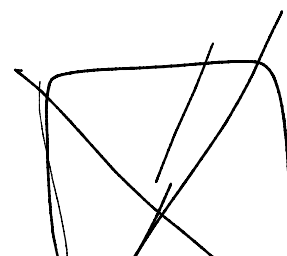


② when we get more and more obs and we decrease the class widths, then number of classes will be increased, and the histogram will be more smoother.

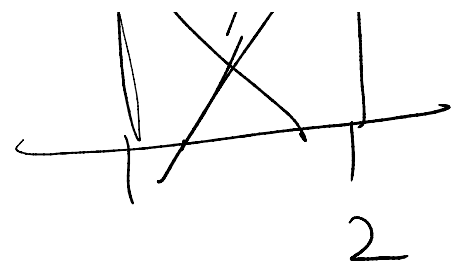
The limiting form of the histogram is called frequency curve.



such that, the

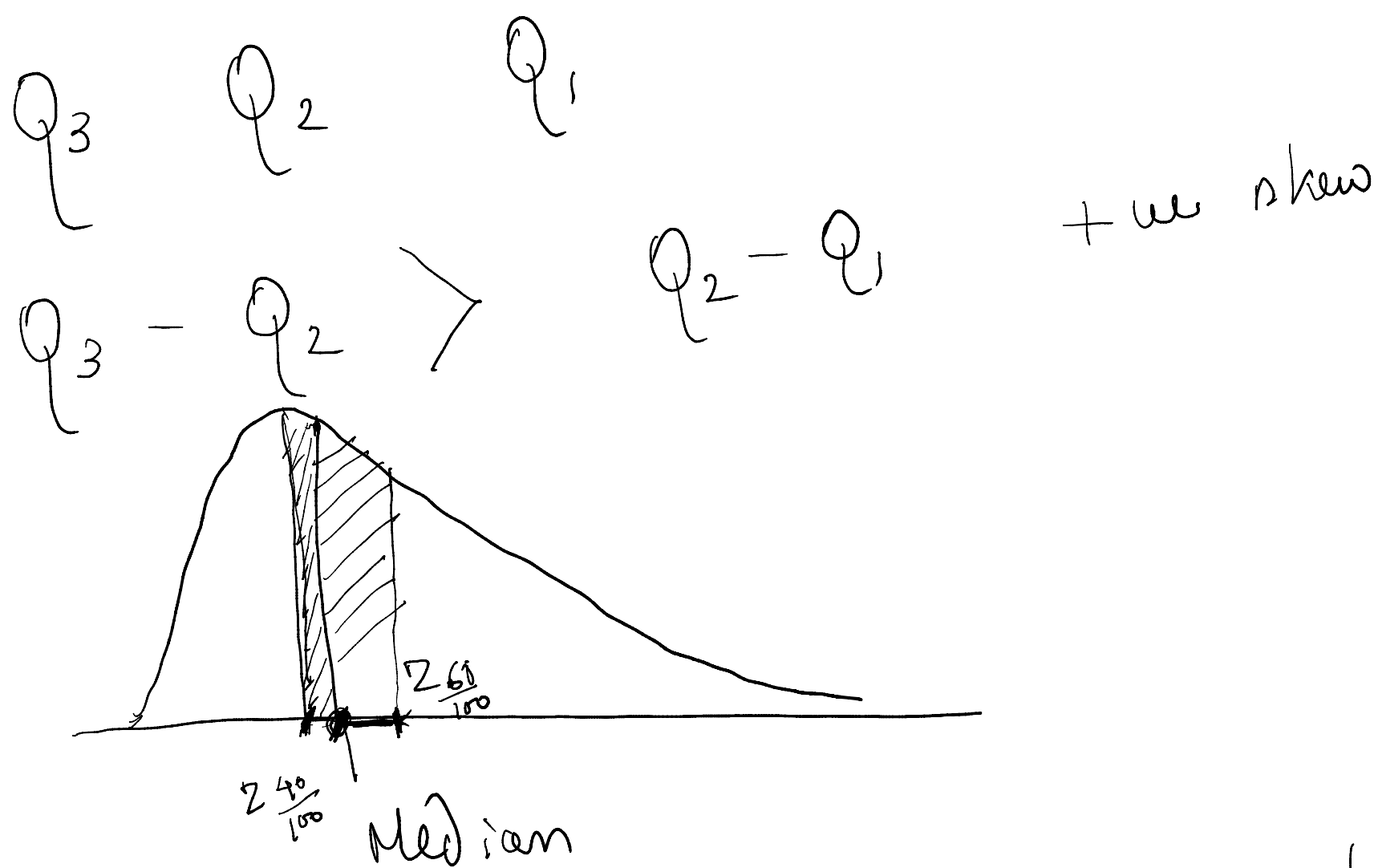


x_{0-h} is x_{0+h}
 If we have a point x_0 such that, the frequency of x_{0-h} and x_{0+h} are same in the frequency dist, then it is symmetrical freq dist. (Pearson's β, γ coefficients)



$$\beta_1 = \frac{m_3^2}{m_2^3}, \quad \beta_1 = 0 \Rightarrow \text{symmetric}, \quad \beta_1 \neq 0 \Rightarrow \text{skew}$$

$$\gamma_1 = \frac{m_3}{m_2^{3/2}}, \quad \gamma_1 = 0 \Rightarrow \text{symmetric}, \quad \gamma_1 > 0 \Rightarrow \text{+ve skew}, \quad \gamma_1 < 0 \Rightarrow \text{-ve skew}$$



$$\begin{aligned}
 Q_3 - Q_2 &< Q_2 - Q_1 && \text{-ve skew} \\
 Q_3 - Q_2 &= Q_2 - Q_1 && \text{Symmetric.}
 \end{aligned}$$

$$sk = \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{(Q_3 - Q_1)}$$

→ Bowley's measurement.